

JAMES BOUTROS

Aspiring Electrical & Computer Engineer | Incoming University of Waterloo ECE, September 2026

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PROFESSIONAL SUMMARY

Disciplined engineering student and dedicated AP student across all four years of secondary school, with hands-on experience building production software across iOS, full-stack web, and embedded systems. Proven ability to take projects from concept to delivery independently, backed by strong mathematical and scientific foundations. Entering the University of Waterloo Electrical and Computer Engineering program in September 2026, with a focus on hardware-software integration and applied problem solving.

EXPERIENCE

iOS Application Developer

Career Education Council / Apple Co-op | Burlington, ON | June 2025 – July 2025

- Designed and developed MicroLoop, a full SwiftUI productivity app featuring micro-habit tracking, AI-powered goal breakdown, streak gamification, push notifications, and a session reflection survey.
- Independently implemented all coded features: dashboard, goal entry flow, animated progress bars, and streak counter.
- Presented the completed application to a panel of Apple engineers at program conclusion.
- Researched emerging iOS frameworks and maintained development documentation for team knowledge transfer.

Peer Tutor

Independent Contract | Burlington, ON | 2025 – 2026

- Delivered 1-on-1 tutoring in Advanced Functions, Grade 12 Chemistry, and Physics to private clients on a self-contracted basis.
- Designed personalized curriculum plans and practice materials tailored to each student's learning style.
- Helped students raise average grades by over 15% through simplified explanations of complex concepts.

Pharmacy Assistant

East Waterdown Pharmacy | Waterdown, ON | July 2023 – September 2023

- Maintained inventory and tracked medication expiration dates to ensure safety and regulatory compliance.
- Assisted pharmacists with prescription preparation, order fulfillment, and counter organization.
- Managed sensitive patient interactions with full professionalism, empathy, and strict confidentiality.

Camp Counsellor

ARSM | Burlington, ON | July 2022 – September 2022

- Supervised children across varying ages and developmental levels, enforcing safety protocols and responding promptly to incidents.
- Led structured group activities and consistently modelled the program's core values.

PROJECTS

MicroLoop | SwiftUI / Xcode / Swift / iOS

Full productivity app built during the CEC Apple co-op. Calendar scheduling, countdown timer, AI goal breakdown, push notifications, gamified streak tracking, and end-of-session reflection survey. Presented to Apple engineers.

JaboGPT | Next.js / React / Gemini 2.5 Flash / Vercel

Production AI chat app at jabogpt.vercel.app. File and image uploads with AI analysis, keyword-searchable sidebar, chat pinning and renaming, and a secure server-side API architecture with zero client-side key exposure.

Personal Portfolio Website | Next.js / React / Framer Motion / Vercel

Built from scratch at jamesboutrosweb.vercel.app. Apple-level scroll animations via Framer Motion, red-black-white design system, single-file content architecture.

TinkerCAD Embedded Systems Suite | C++ / Arduino Uno / TinkerCAD

Four complete hardware systems: dual-sensor fire alarm (MQ gas + TMP), PIN-based two-door lock (4x4 keypad, servo, I2C LCD), ISR-based pedestrian traffic light, and photoresistor ADC ambient light classifier with LED zone mapping.

OpenProcessing Mini-Projects | p5.js / C++ / JS / HTML / CSS

Interactive mini-projects on OpenProcessing including a stock market dashboard, an engineering university information interface, and a neuro-evolution (ML) driven mini-game. openprocessing.org/@u583961

References available upon request

EDUCATION

University of Waterloo

BASc, Electrical and Computer Engineering
Starting September 2026

Assumption Catholic S.S.

Ontario Secondary School Diploma
Burlington, ON | Graduating 2026

98% Avg | 4.0 Unweighted GPA

AP courses: Advanced Functions, Calculus AB (Score: 5), Physics 1, Chemistry, English
Self-studying: AP Calculus BC, AP Physics C E&M

TECHNICAL SKILLS

Languages

Swift, Python, C++, JavaScript, HTML, CSS

Frameworks

SwiftUI, Next.js, React, Framer Motion

Hardware

Arduino Uno, TinkerCAD, ADC, servo motors, ISR, I2C LCD

Developer Tools

Xcode, VS Code, Cursor, OpenAI Codex CLI, Antigravity IDE, Git, GitHub, Vercel, Supabase

AI & Tooling

MCP integrations, Claude API, agentic coding workflows, prompt engineering, terminal-first development

Concepts

OOP, interrupt-driven I/O, state machines, REST APIs, analog sensing

ACCOMPLISHMENTS

- AP Calculus AB Exam — Score of 5
- Galois Waterloo Math Contest — Top Score at School
- Grade 11 AP Advanced Functions Course Award
- Honor Roll — All 4 Years (2023–2026)

EXTRACURRICULAR ACTIVITIES

- Peer Tutoring — Math, Chemistry, Physics
- Model UN — In-House Conference (Gr. 12)
- DECA — Regional Competition, Brock University (Gr. 12)
- VEX Robotics Club (Gr. 10–12)
- Senior Reach (Gr. 11)
- Competitive Weightlifting, 3 Years
- Chess Club (Gr. 9–10)
- 160+ Community Volunteer Hours

CERTIFICATIONS

- First Aid and CPR Certified
- Babysitting Certified
- ICT SHSM Certifications

ONLINE PRESENCE

Portfolio: jamesboutrosweb.vercel.app

JaboGPT: jabogpt.vercel.app

GitHub: github.com/jcb1515

LinkedIn: linkedin.com/in/james-b-2682403b3